VF Asia Footwear Laboratory Testing Manual

Test Method for Whole Shoe Water Absorption	Test method # VF- G06 Page 1 of 2
Date Issued : May,2019 Revision # : 1 Reason for revision : On the report , add the condition of liner and sock-liner , indicate if wet or dry.	Created by : Nila Babia Erazo

PURPOSE AND SCOPE: To measure the amount of water absorbed by a shoe as a result of being immersed in water for a specified period of time.

REFERENCE:

EQUIPMENT AND MATERIAL:

1. Immersion tank which is large enough to contain the test shoe such that no portion of the test shoe contacts the sides of the tank.
2. A balance capable of weighing to 5000 grams +/-0.010grams.
3. Whole Shoe
4. Steel scale capable of measuring to +/-0.1cm.
5. Water at normal temperature

SAMPLE PREPARATION:

1. Condition the shoe sample for at least 48 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity to allow the cement to crosslink completely.
2. Visually check any bond gaps between the upper and bottoms. Any sign of bond gap, the shoe sample should be discarded.

REPETITIONS: Test two (2) shoe samples**.**

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

1. Weigh the test shoe sample to the nearest 0.1 g (W1).
2. Place the test shoe sample in the immersion tank.
3. Carefully pour water at 23+/-2ºC into the immersion tank. Waterline is 20 mm above the top line of outsole sidewall. Ensure that the water level must be below the vamp.

Test Method for Whole Shoe Water Absorption	Test method # VF- G06 Page 2 of 2
Date Issued : May,2019 **Revision # : 1** **Reason for revision : On the report , add the condition of liner and sock-liner , indicate if wet or dry.**	**Created by : Nila Babia Erazo**

4. If needed, insert enough weight to hold the shoe sample firmly on the bottom of the container.
5. Allow the test shoe sample to remain immersed in water for 30 minutes.
6. Carefully remove the test shoe sample from the water, use caution not to drip water into footwear. Lightly blot the surface of the sample to remove any surface water.
7. Immediately weigh the test shoe material to the nearest 0.10 g (W2).
8. Compute water absorbed
9. Check for any leakage area. Check the inside of liner and sock-liner if wet or any presence of water inside. Stop the testing when leaks are detected.
10. If no leaks are detected, continue testing. Repeat steps 2 to 7
11. If the inside of liner and sock-liner are still dry after one hour. It means it has PASS the testing.

CALCULATION:

Water absorption (%): $= \frac{W2 - W1}{W1} * 100$

REPORT:

1. The average of the two measurements to the nearest 0.1%.
2. Condition of inside of liner and sock-liner : wet or dry